UNIVERSITY of WASHINGTON

UW Math AI Lab

Spring 2026 · Provable Computation

VERIFIED EXECUTABLE LINEAR ALGEBRA

DenseMatrix makes Lean matrix algorithms run like data.

Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly Matrix theorems.

TEAM

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SCAN FOR ARTIFACT

Why DenseMatrix

Row-major arrays let kernels work over contiguous data instead of re-evaluating pointwise Matrix functions.

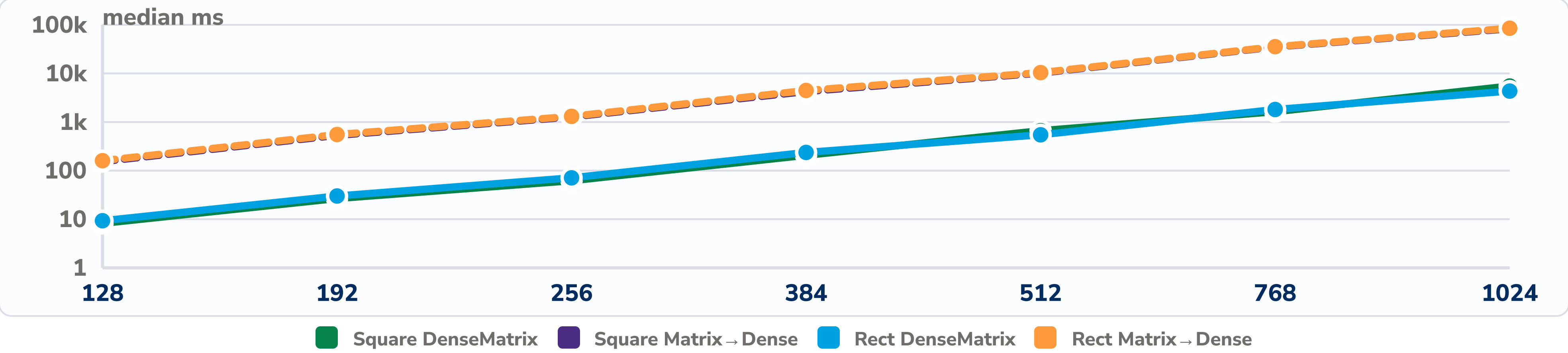
Matrix API → DenseMatrix.ofMatrix → Row-major kernels

Materialized executable workloads only; Matrix remains the proof API.

Multiplication Scaling

15.1-21.5x faster

DenseMatrix vs Matrix→Dense, multiplication median ms.



Benchmark Scope

2247 RECORDS	**328** MATERIALIZED CASES	**2048** MAX EXTENT

Protocol	30 repeats · 5 warmups
Sizes	16, 32, 64, 96, 128, 192, 256
Stress	384, 512, 768, 1024
Machine	pinned CPU core · archived environment

Proof Connection

- `DenseMatrix.toMatrix_add`
- `DenseMatrix.toMatrix_mul`
- `rowEchelonForm_toMatrix`
- `luDet_eq_det`
- `gaussDet_eq_det`

Theorems connect executable kernels back to the Matrix API.

Representation Tradeoff

Median ms for pointwise primitives at n=1024.

OPERATION	DENSE	MATRIX	MATRIX→DENSE	FASTEST
Add	54.0	45.1	47.5	Matrix
Scalar	53.2	30.3	35.3	Matrix
Transpose	55.3	43.5	46.8	Matrix

Algorithm Kernels

Representative rational kernels at n=128.

REF	16,292 ms
RREF	32,394 ms
LU factor	17,090 ms
Gauss det	16,234 ms
LU det	16,933 ms